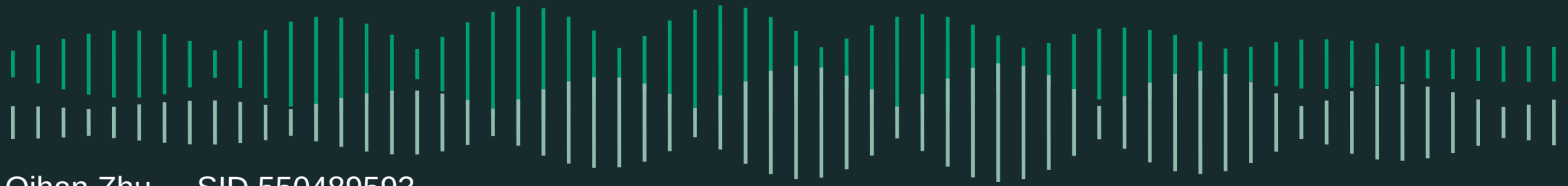
WHEN BACKGROUND NOISE HAS MEANING

Environmental sound classification under semantic interference



Qihan Zhu · SID 550489593

Same target. Same relative level. Different interference.

2,000 ESC-50 clips · official five-fold evaluation · paired mixtures for every model

White noise

Acoustically nonspecific

DEMAND

Traffic · station · metro

Same category

Different fine class

Different category

Different coarse category

+10 dB

Weaker interference

+5 dB

Primary comparison

0 dB

Equal-power stress test

24,000 own-activity mixtures; 23,991 target-activity mixtures after three undefined pairs × three levels are excluded.

Three representations, one controlled experiment

Deployment behaviour is comparable; total pretraining history is not.

A **Handcrafted + SVM**

Training-fold standardisation

STFT → spectral descriptors
MFCCs + deltas → 92 features

B **Compact log-Mel CNN**

Three optimisation seeds

STFT → Mel filters → log power
36,258 trainable parameters

C **Frozen LAION-CLAP**

No training or fine-tuning

Audio / text embedding similarity
50 fixed class descriptions

Custom STFT checked against SciPy and actual course MATLAB code; maximum absolute discrepancy 2.93×10^{-14}.

Power matching depends on where power is measured

The original target-mask definition is retained as a full sensitivity analysis.



Own-activity power

Measure each source over its own active interval.

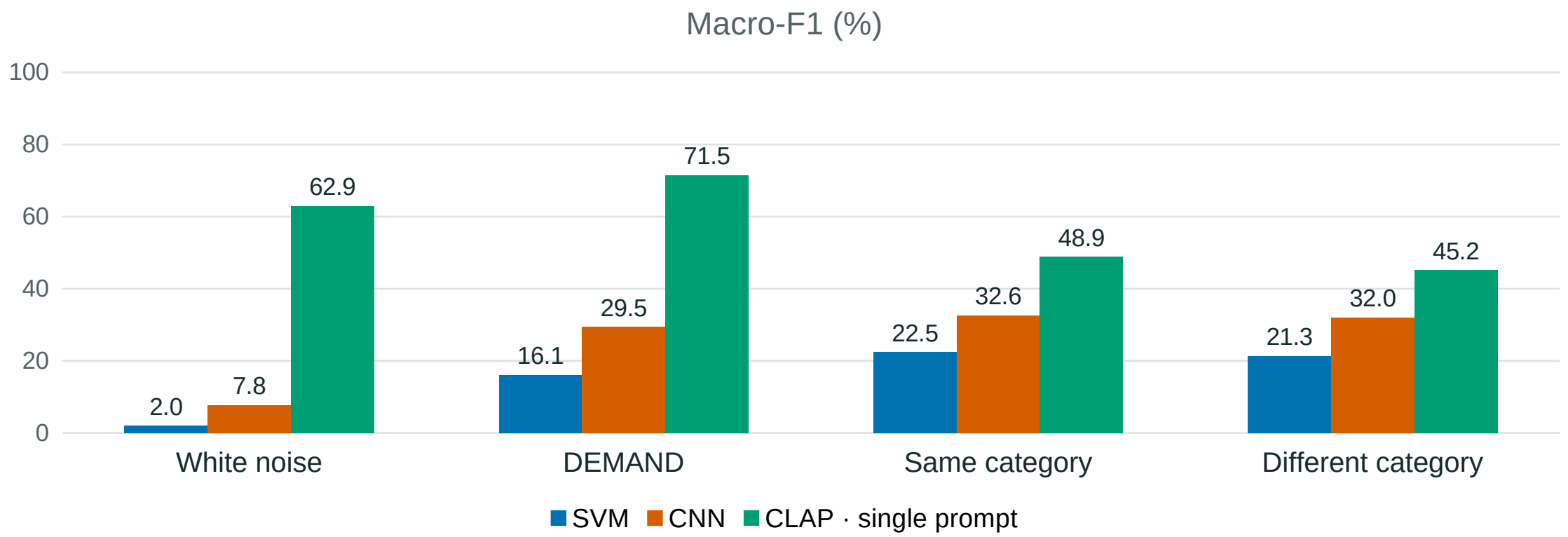
Target-activity power

Measure both sources over the target’s active interval.

Illustrative masks. Common peak attenuation prevents clipping; a gain-matched clean control isolates its effect.

Interference power alone does not predict failure

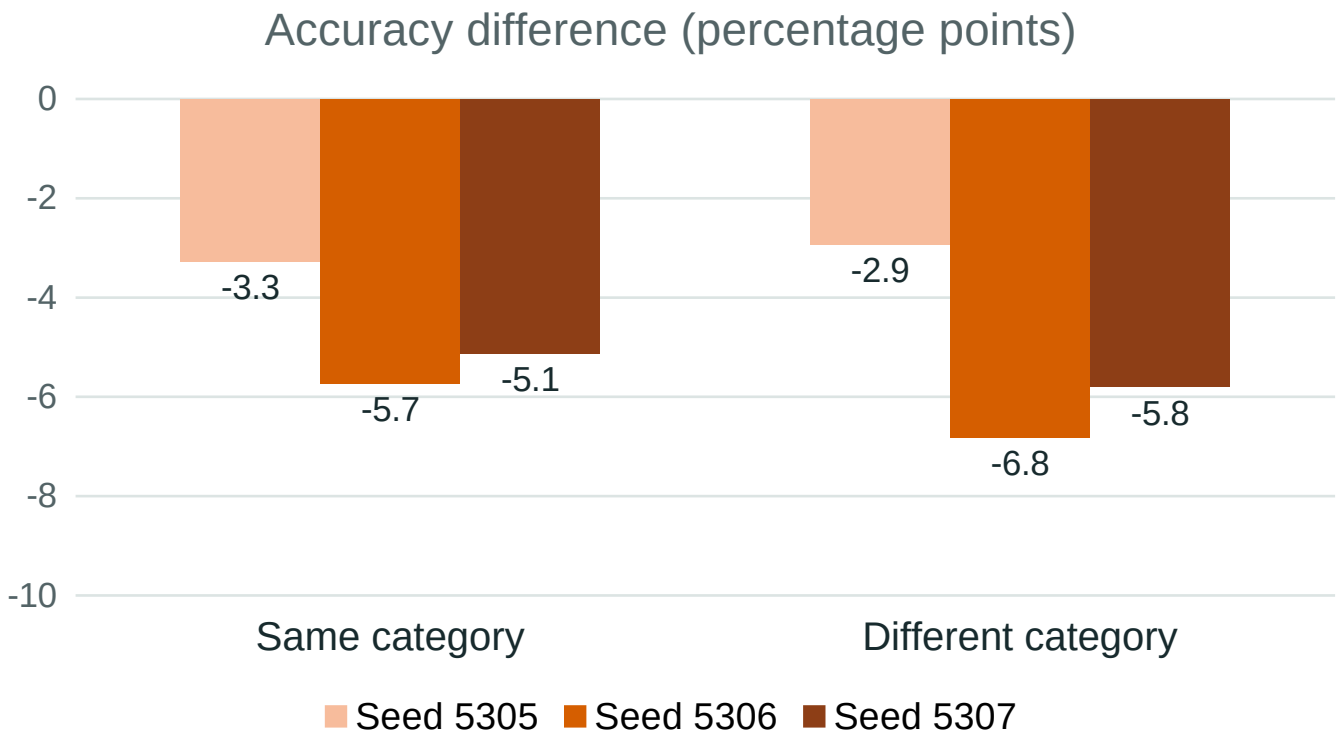
Own-activity protocol · +5 dB · five-fold mean macro-F1



The report gives fold SD, all three levels, retained-target comparisons and the complete target-activity counterpart.

The CNN overlap contrast replicates across seeds

Semantic minus matched DEMAND accuracy at +5 dB, overlap ≥ 0.8



Direction repeats

All three seeds favour DEMAND on these matched targets.

The two new seeds pass the declared
Holm family. Seed 5305 does not.

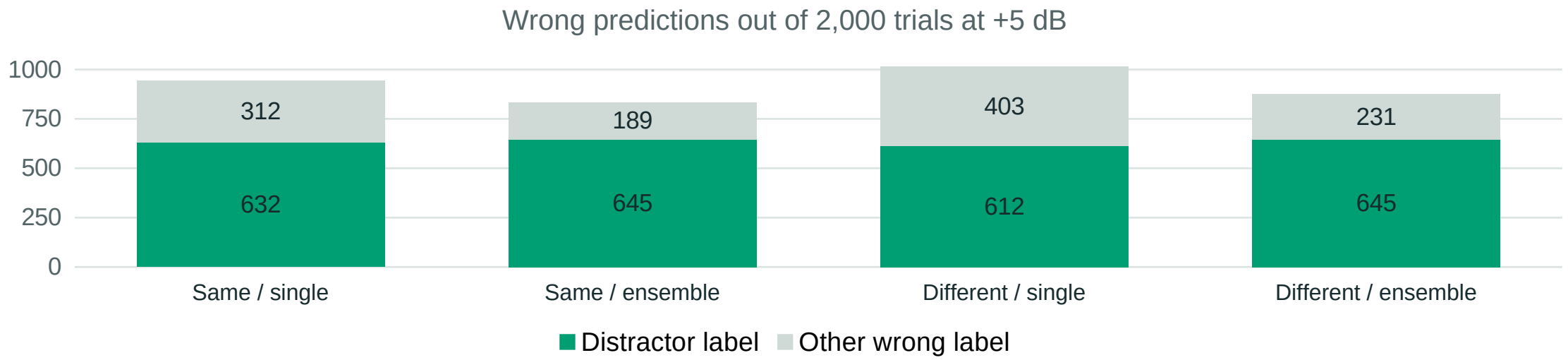
Report: per-seed paired 95% intervals and all 12 corrected tests. Repeated predictions of one target are not independent samples.

Better class readout still leaves distractor competition

Frozen audio encoder · all 300 released prompt entries · no test-label prompt selection

81.75% → 92.90%

Clean accuracy: single prompt → fixed ensemble



This is a prompt-ensemble ablation, not full PAT. Deduplicating the 275 unique templates changes F1 by at most 0.47 pp.

Does robustness training transfer to competing events?

Same augmentation probability and epoch cap · clean validation · three CNN seeds

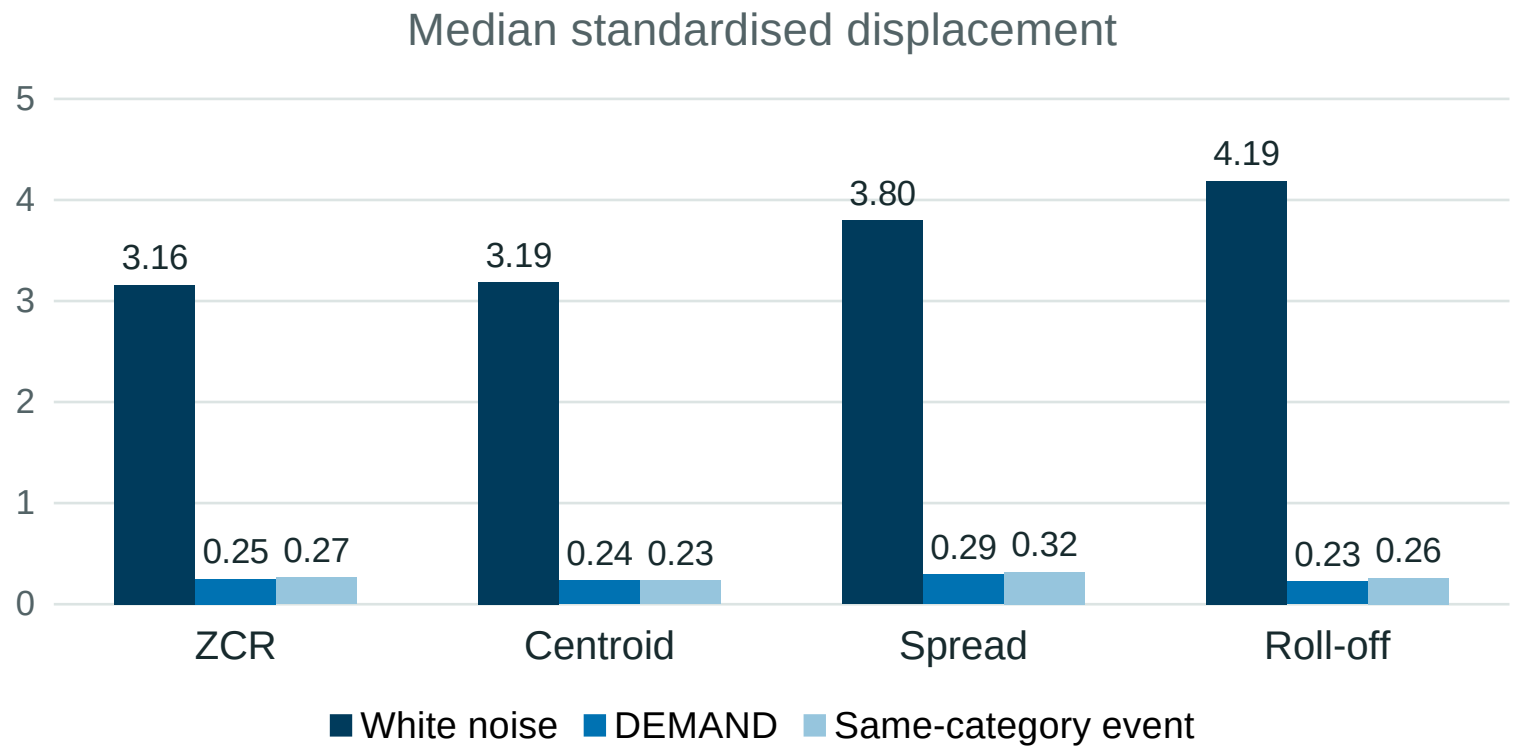
Model / training	Clean	White	DEMAND	Same	Different
SVM / clean	48.0 ± 4.1	2.0 ± 0.9	16.1 ± 2.0	22.5 ± 2.7	21.3 ± 2.6
SVM / WGN	46.0 ± 4.1	41.1 ± 5.4	23.7 ± 5.1	22.6 ± 3.4	22.2 ± 2.6
SVM / semantic	49.3 ± 2.4	1.9 ± 0.6	19.9 ± 2.6	26.8 ± 2.3	25.0 ± 0.9
CNN / clean	65.8 ± 0.7	7.5 ± 0.4	30.9 ± 1.4	33.0 ± 0.3	31.4 ± 0.5
CNN / WGN	64.6 ± 0.6	43.9 ± 2.7	33.8 ± 2.2	33.5 ± 1.0	32.1 ± 1.1
CNN / semantic	61.7 ± 2.4	9.9 ± 0.2	33.6 ± 1.6	36.1 ± 0.8	33.7 ± 0.9

Own activity · macro-F1 (%) · interference conditions at +5 dB

CNN: mean ± seed SD; SVM: mean ± fold SD. Direct CNN semantic-over-WGN accuracy gains do not pass the 156-test Holm correction.

The SVM weakness is visible in its DSP features

Standardised within-model displacement from the gain-matched clean control



White noise shifts spectral shape

Movement alone is insufficient:
its direction relative to the classifier also matters.

Training-fold scales; each group divided by √dimension. CNN and CLAP cosine distances are interpreted in separate spaces.

Listen to a decision being captured

Can opening + clock alarm · +5 dB SIR · full target-activity overlap

Gain-matched clean control

Play 5-second audio

Target + weaker clock alarm

Play 5-second audio

SVM

can opening
↓
chirping birds

CNN

can opening
↓
clock alarm

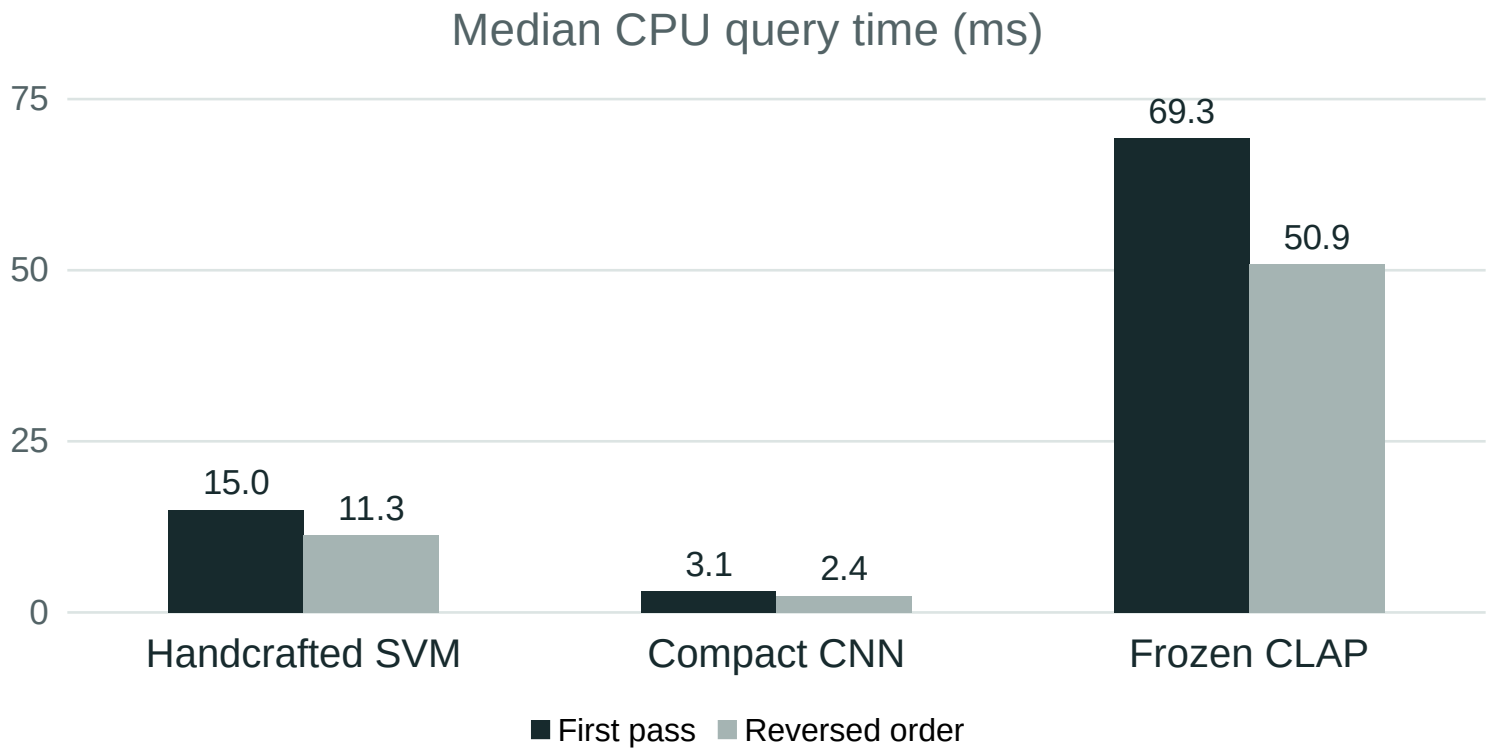
CLAP

can opening
↓
clock alarm

PPTX: embedded audio; PDF: linked audio. Illustrative case using core CNN seed 5305 and single-prompt CLAP; provenance in notes.

Cost ordering survives timing drift

Same machine · CPU · batch size one · preprocessing and class scoring included



36,258

CNN parameters

158.3 million

CLAP parameters

250 clips × three repetitions per pass. The reversed-order pass checks observed drift; both are retained. Report includes IQR and p95.

Power is only part of the story.

Controlled mixtures reveal where the classification decision moves.

01 **Representation matters**

White noise and meaningful events produce different failure patterns.

02 **Controls change interpretation**

Activity masks, temporal overlap and matched clean controls must be explicit.

03 **Robustness needs diagnosis**

Report error destinations, transfer and uncertainty alongside accuracy.

Limits: three backgrounds · one pair per category type · pretraining overlap · 0-dB label ambiguity

Evidence, code and reproducible materials

Editable figures · numerical tables · manifests · model hashes · attributed listening examples

ESC-50 / Piczak	Official folds, dataset caveats and classic spectrogram CNN
LAION-CLAP / Wu et al.	Frozen audio-language reference, ICASSP 2023
PAT / Seth et al.	Released prompt-list ablation; full PAT is not reproduced
Mix-CLAP / Kohno et al.	Noise-aware audio-language model context, ICASSP 2026
RSA-Bench / Zhang et al.	Structured-background speech/reasoning context, ACL Findings 2026

github.com/Charlie-Uni/elec5305-project-550489593

Audio: LG, Can Open.WAV (CC BY); hookhead, alarm clock.aif (CC0). ESC-50 compilation: CC BY-NC 3.0.

Draft for author review. AI assistance was used for implementation, checking and preparation; confirm course disclosure and video format.